Testing the Huber weights in space (E. THEBAULT)

In this report, I estimate the maximum likelihood model computed in space from all candidate models using an iterative weighting scheme. The Gauss coefficients of each candidate model are used to compute the vector field at the Earth's mean radius on the node of a quasi-regular grid in space. The field is computed on 10000 points on the sphere. For each node we obtain as many values as the number of candidate models depending on the considered epoch. The algorithm then reweights each model automatically in order to highlight the geographical regions where models agree and disagree. This analysis complements the cross-analysis between models performed by different teams. It could serve as a guide for voting which weighting scheme can be used for the production of IGRF-13. The major advantage of this approach is that all models are considered: Even if some disagree in some regions, they are informative in some others.

The Gauss coefficients of the Huber weighted model are provided in each section for further comparisons. In addition, the Huber model for the DGRF 2015 is compared to the ASM scalar data centered on 1st January 2015 +/- 15 days. For DGRF, IGRF and SV, the Huber weighted model is also compared to the median model computed directly from the Gauss coefficients of the candidate models. This should help estimating if the median and the Huber models really differ.

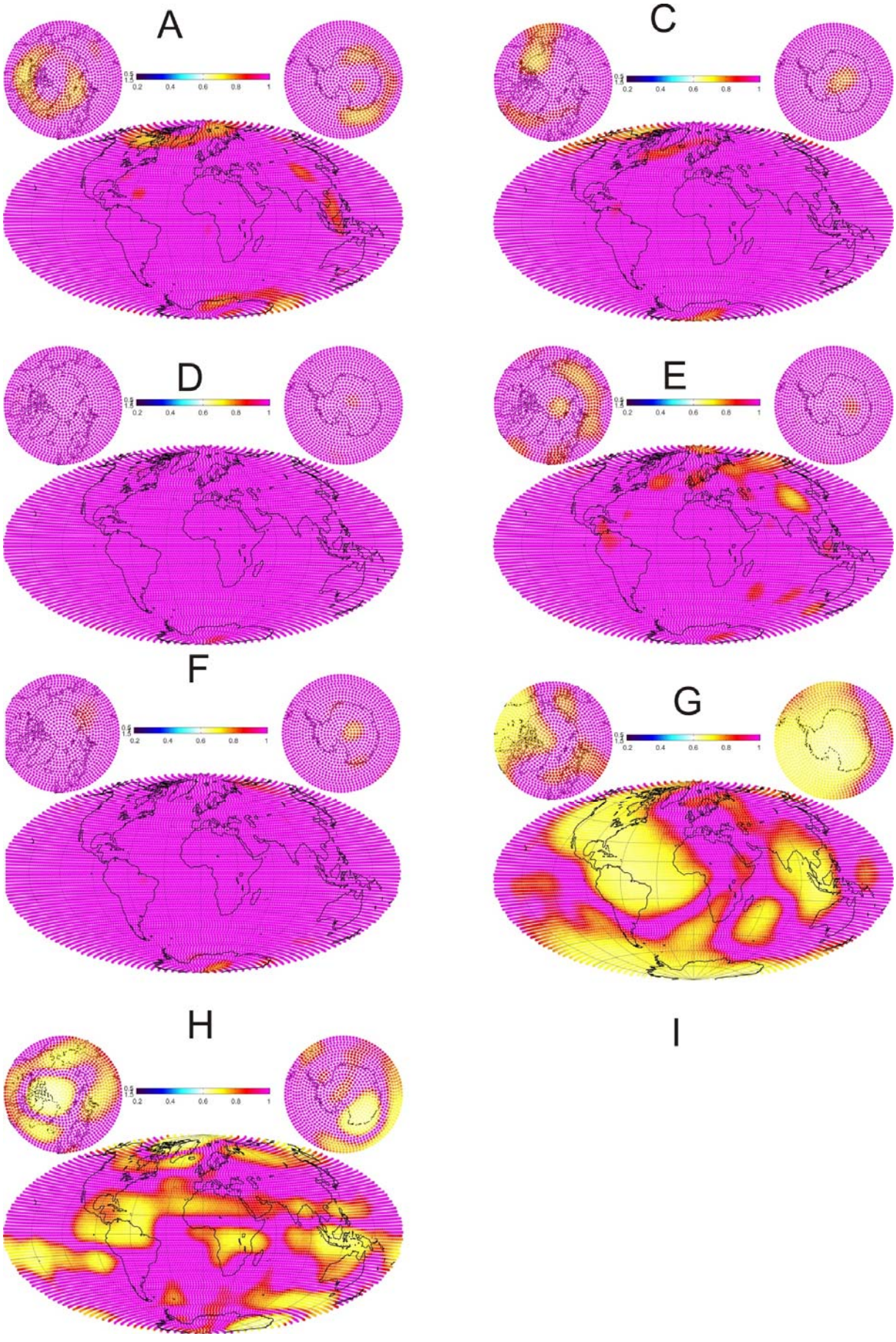
In each section, I first display the weights of each candidate models for the radial component only estimated iteratively. The coding for each institution is

Table 1: Coding for for the institution team

A	BGS candidates
B	CEA_CSES candidates
C	CU_NCEI candidates
D	DTU candidates
E	GFZ candidates
F	IPGP candidates
G	ISTerre candidates
H	IZMIRAN candidates
I	Japanese Team candidates
J	Leeds candidates
K	Max Plank candidates
L	NASA_GSFC candidates
M	Potsdam_MaxPlank candidates
N	Spanish Team candidates
O	Strasbourg candidates

DGRF 2015

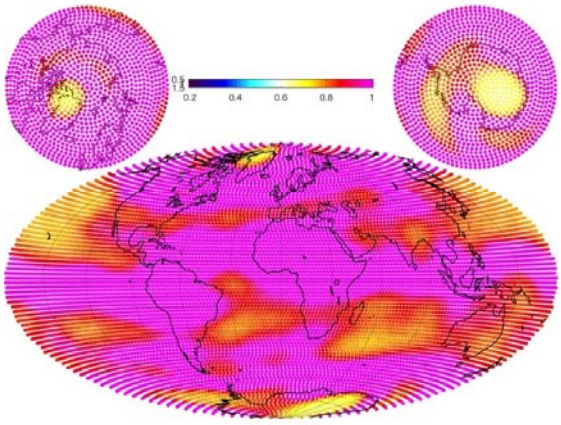
Three models, G, H and O are significantly down weighted. Two models L and N are moderately down weighted. However, all candidate models agree in space in many regions. The Huber weighted model and the median model are in excellent agreement with RMS difference in excess of 2-3 nT at the Earth's mean radius. **If we choose simplicity, the median model could be used for DGRF 2015.**


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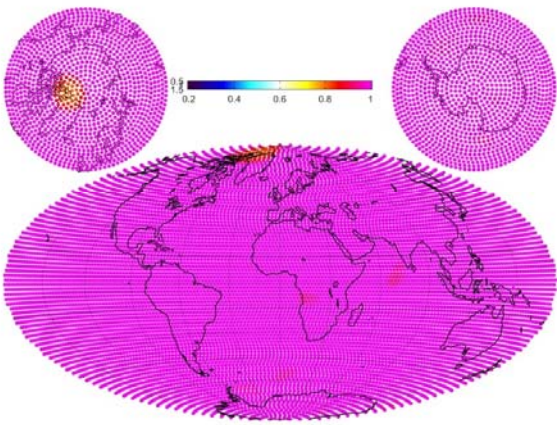
J

K

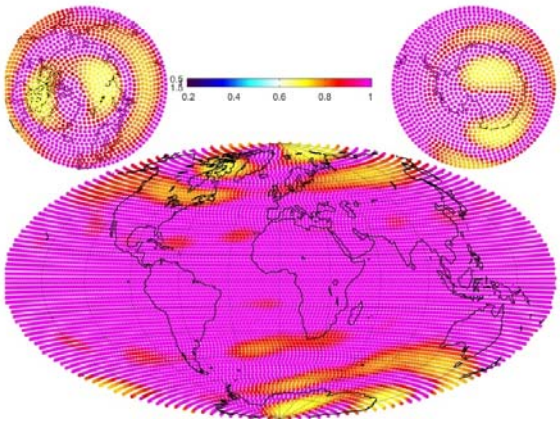
L



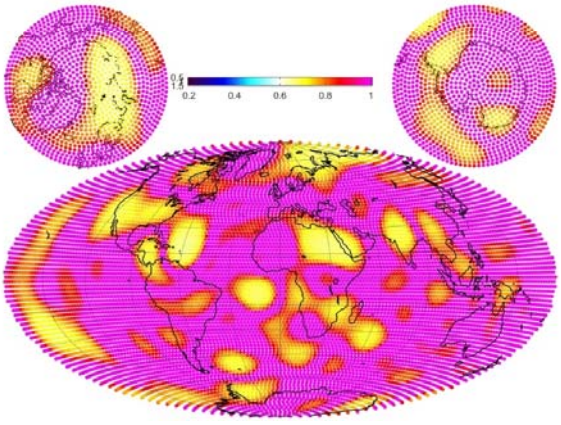
M



N

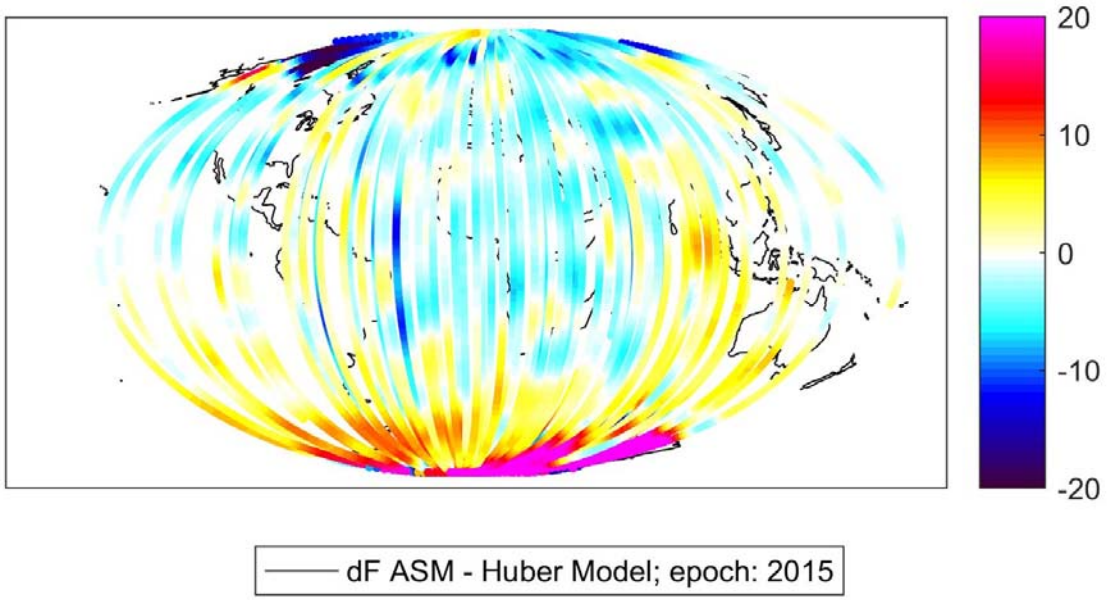


O

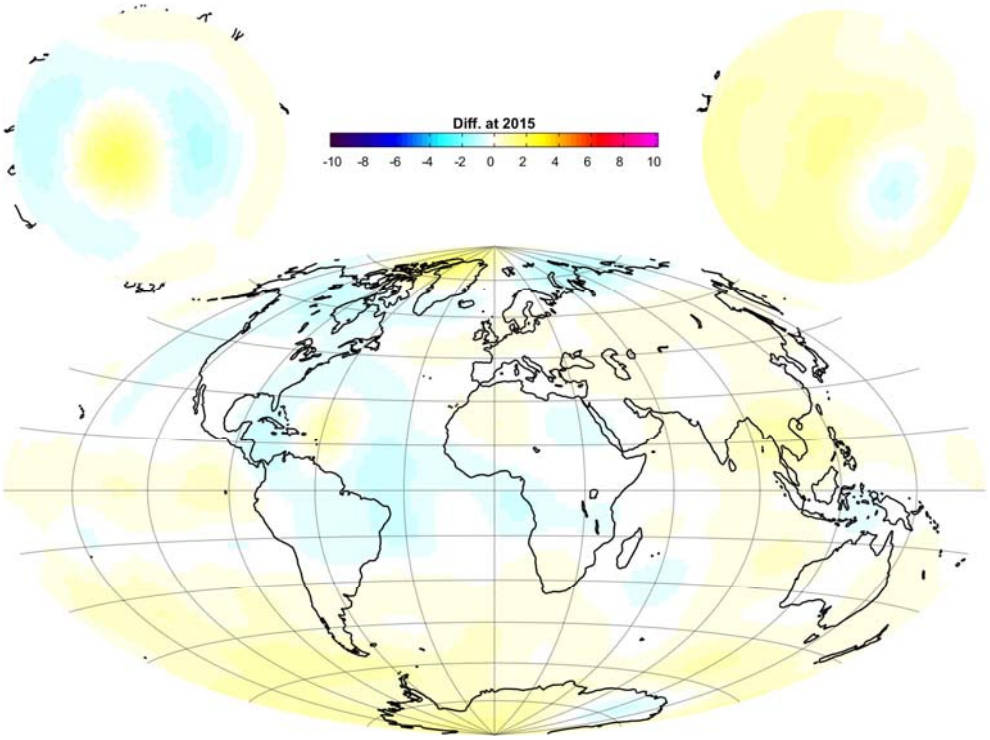


Comparison of Huber weighted model with the ASM data and the median model

The residuals between the scalar ASM measurements, corrected for the magnetospheric and the crustal fields, and the Huber weighted model are shown below. Structures are apparent in the polar regions as a result of strong external fields but no organized residuals are otherwise prominent.



The difference between the Huber weighted model and the median model show no significant or strong differences at the Earth's mean radius. This could be a way to consider the median model as a good candidate for DGRF 2015.0.



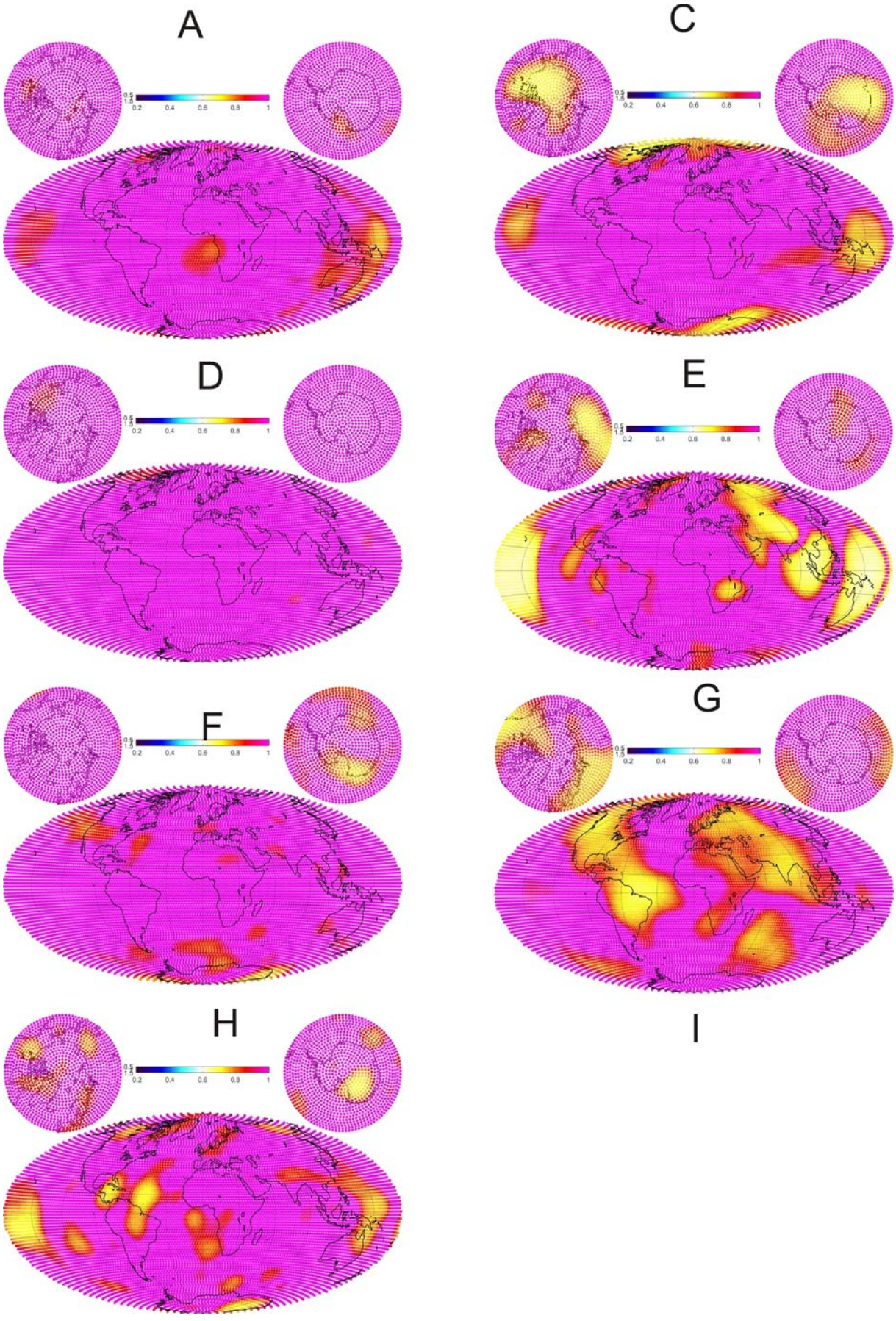
Huber weights DGRF 2015 model coefficients:

```
Candidate model with Huber weights in space, E. Thebault, 15-Nov-
2019 13:02:46
Epoch = 2015.00
n   m        gnm        hnm
  1   0  -29441.71       0
  1   1   -1501.78    4796.08
  2   0   -2445.91       0
  2   1    3012.19   -2845.35
  2   2    1676.37    -642.16
  3   0    1350.40       0
  3   1   -2352.28    -115.25
  3   2    1225.82     245.03
  3   3     581.71    -538.77
  4   0     907.40       0
  4   1     813.65     283.57
  4   2     120.49    -188.44
  4   3    -334.87     180.95
  4   4      70.37    -329.21
  5   0    -232.82       0
  5   1     360.16      46.94
  5   2     192.35     196.96
  5   3    -140.93    -119.15
  5   4    -157.40      15.97
  5   5       4.25     100.11
  6   0      69.53       0
  6   1      67.56     -20.60
  6   2      72.81      33.30
  6   3    -129.86      58.73
  6   4     -28.92     -66.66
  6   5      13.14       7.36
  6   6     -70.85      62.41
  7   0      81.28       0
  7   1     -75.98     -54.25
  7   2      -6.78     -19.51
  7   3      51.82       5.58
  7   4      15.06      24.45
  7   5       9.33       3.27
  7   6      -2.86     -27.51
  7   7       6.61      -2.33
  8   0      23.99       0
  8   1       8.89      10.02
  8   2     -16.78     -18.26
  8   3      -3.15      13.20
  8   4     -20.57     -14.60
  8   5      13.33      16.17
  8   6      11.76       5.68
  8   7     -15.97      -9.10
  8   8      -2.02       2.26
  9   0       5.32       0
  9   1       8.84     -21.78
  9   2       3.03      10.75
  9   3      -3.22      11.74
  9   4       0.67      -6.75
  9   5     -13.21      -6.88
```

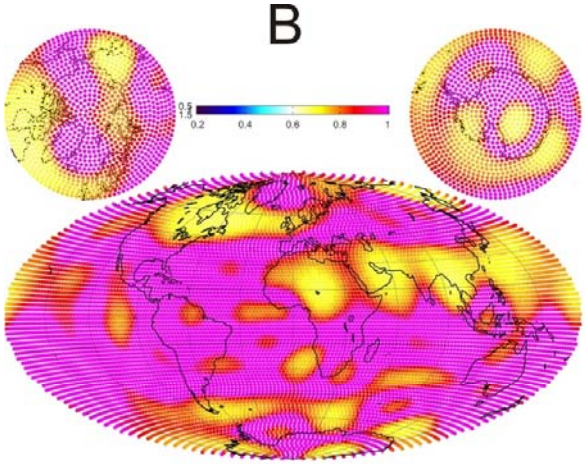
9	6	-0.10	7.79
9	7	8.70	1.05
9	8	-9.07	-3.88
9	9	-10.54	8.44
10	0	-2.01	0
10	1	-6.27	3.27
10	2	0.16	-0.40
10	3	0.53	4.55
10	4	-0.54	4.41
10	5	1.71	-7.92
10	6	-0.68	-0.61
10	7	2.13	-4.16
10	8	2.32	-2.84
10	9	-1.80	-1.10
10	10	-3.59	-8.72
11	0	2.98	0
11	1	-1.39	0.01
11	2	-2.29	2.10
11	3	2.08	-0.59
11	4	-0.79	-1.06
11	5	0.58	0.76
11	6	-0.70	-0.20
11	7	0.13	-2.12
11	8	1.70	-1.44
11	9	-0.22	-2.57
11	10	0.44	-2.01
11	11	3.49	-2.35
12	0	-2.07	0
12	1	-0.16	-1.12
12	2	0.44	0.38
12	3	1.23	1.76
12	4	-0.89	-2.18
12	5	0.85	0.27
12	6	0.11	0.72
12	7	0.54	-0.08
12	8	-0.37	0.30
12	9	-0.43	0.23
12	10	0.22	-0.90
12	11	-0.94	-0.16
12	12	-0.03	0.72
13	0	-0.02	0
13	1	-0.90	-0.86
13	2	0.44	0.46
13	3	0.61	1.55
13	4	-0.42	-0.50
13	5	0.96	-1.23
13	6	-0.19	-0.10
13	7	0.81	0.42
13	8	-0.13	-0.04
13	9	0.38	0.47
13	10	0.09	0.47
13	11	0.47	-0.30
13	12	-0.35	-0.43
13	13	-0.36	-0.71

IGRF-2020

For IGRF 2020, candidate models that differ the most are B, E, N, and O. These models show significant downweighted values down to 0.6 in some regions. However, we notice that models G and L have large structures moderatly downweighted in space. **Differences in space between the median and the Huber model are small and in excess of 3-4 nT. Therefore, the median model could be used for simplicity.**

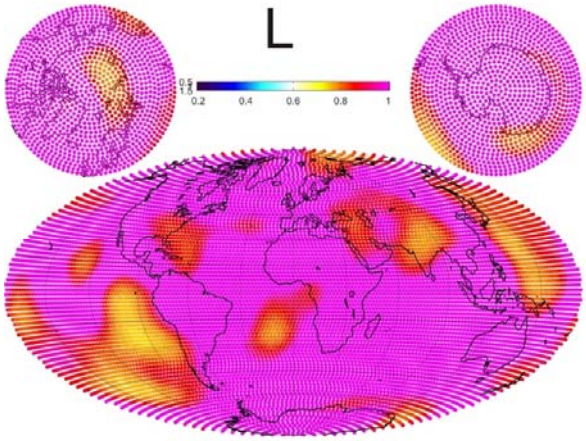

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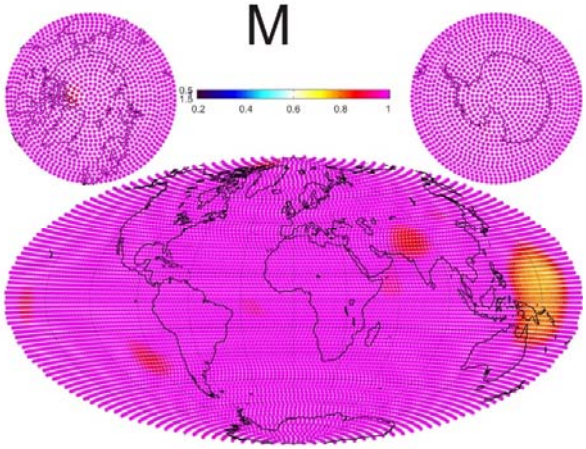


K

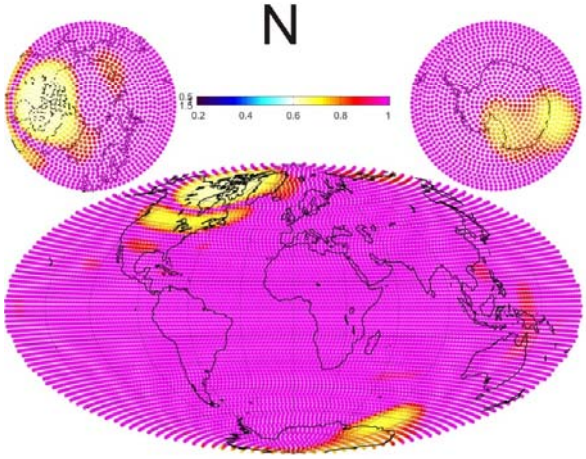
L



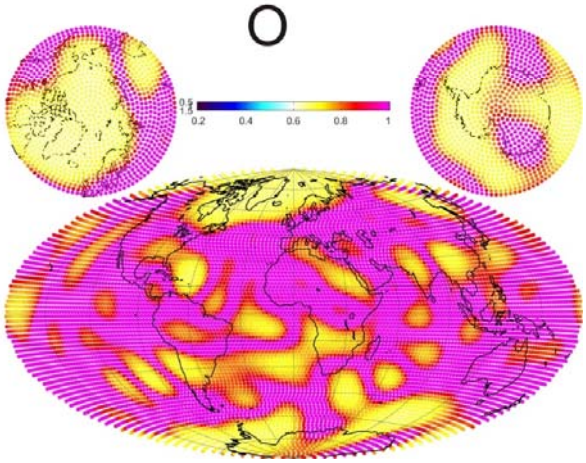
M



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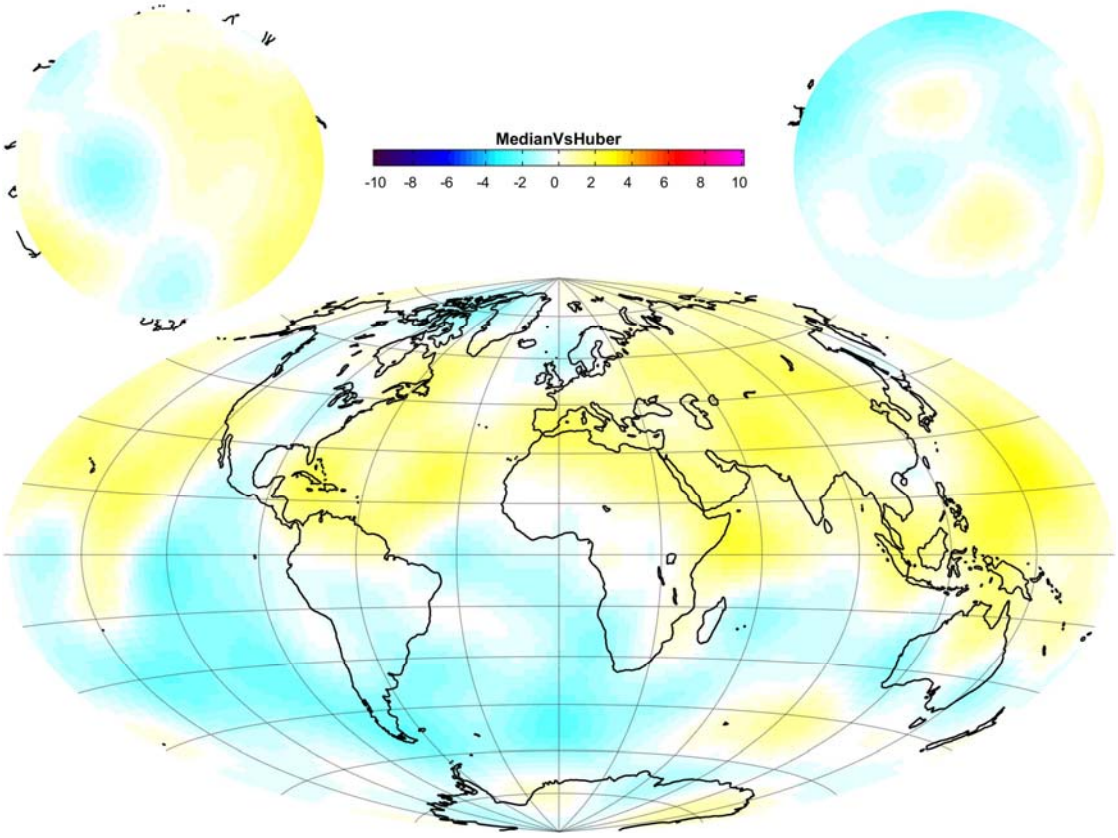
Huber weighted Gauss coefficient for IGRF 2020

Candidate model with Huber weights in space, E. Thebault, 15-Nov-2019 13:29:46
Epoch = 2015.00

n	m	gnm	hnm
1	0	-29405.24	0
1	1	-1450.86	4652.24
2	0	-2499.55	0
2	1	2981.93	-2991.55
2	2	1676.93	-734.48
3	0	1363.49	0
3	1	-2381.21	-82.11
3	2	1236.20	241.83
3	3	525.85	-543.45
4	0	902.96	0
4	1	809.46	281.90
4	2	86.32	-158.46
4	3	-309.40	199.71
4	4	48.12	-349.75
5	0	-234.37	0
5	1	363.28	47.64
5	2	187.85	208.37
5	3	-140.64	-121.16
5	4	-151.21	32.28
5	5	13.48	98.84
6	0	66.01	0
6	1	65.57	-19.13
6	2	72.93	25.14
6	3	-121.53	52.74
6	4	-36.24	-64.44
6	5	13.51	8.90
6	6	-64.68	68.13
7	0	80.60	0
7	1	-76.68	-51.42
7	2	-8.22	-16.87
7	3	56.47	2.23
7	4	15.75	23.47
7	5	6.39	-2.20
7	6	-7.23	-27.23
7	7	9.80	-1.85
8	0	23.70	0
8	1	9.76	8.34
8	2	-17.60	-15.25
8	3	-0.42	12.83
8	4	-21.05	-11.77
8	5	15.34	14.93
8	6	13.65	3.60
8	7	-16.56	-6.89
8	8	-0.35	2.81
9	0	5.02	0
9	1	8.36	-23.41
9	2	2.87	10.98
9	3	-1.44	9.82
9	4	-1.12	-5.12
9	5	-13.22	-6.27

9	6	1.08	7.80
9	7	8.87	0.42
9	8	-9.28	-1.47
9	9	-11.87	9.62
10	0	-1.86	0
10	1	-6.24	3.39
10	2	-0.14	-0.17
10	3	1.70	3.56
10	4	-0.84	4.85
10	5	0.67	-8.61
10	6	-0.94	-0.08
10	7	1.88	-4.27
10	8	1.45	-3.45
10	9	-2.38	-0.08
10	10	-3.85	-8.75
11	0	2.97	0
11	1	-1.36	-0.00
11	2	-2.50	2.48
11	3	2.31	-0.56
11	4	-0.92	-0.39
11	5	0.31	0.64
11	6	-0.72	-0.21
11	7	-0.12	-1.67
11	8	1.45	-1.58
11	9	-0.63	-3.01
11	10	0.20	-1.98
11	11	3.10	-2.55
12	0	-1.96	0
12	1	-0.13	-1.19
12	2	0.42	0.55
12	3	1.28	1.39
12	4	-1.15	-1.79
12	5	0.73	0.06
12	6	0.33	0.75
12	7	0.49	-0.14
12	8	-0.25	0.56
12	9	-0.48	0.17
12	10	0.11	-0.93
12	11	-1.12	-0.00
12	12	-0.27	0.53
13	0	0.05	0
13	1	-0.96	-0.90
13	2	0.52	0.62
13	3	0.73	1.38
13	4	-0.31	-0.38
13	5	0.77	-1.32
13	6	-0.01	-0.09
13	7	0.77	0.30
13	8	-0.05	-0.09
13	9	0.39	0.47
13	10	0.12	0.50
13	11	0.47	-0.37
13	12	-0.47	-0.35
13	13	-0.41	-0.60

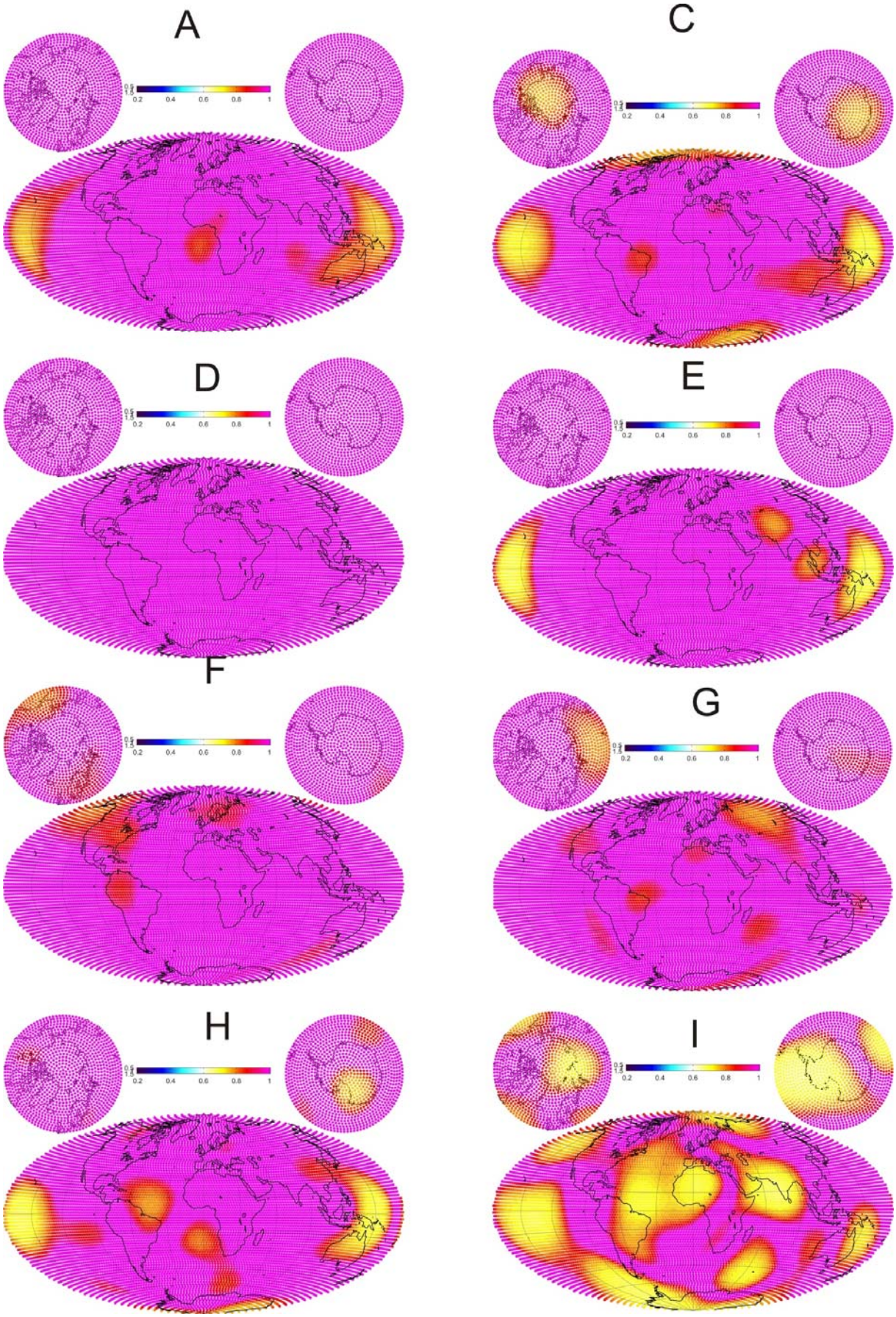
Difference between the IGRF 2020 median and the Huber models at the Earth's mean radius.

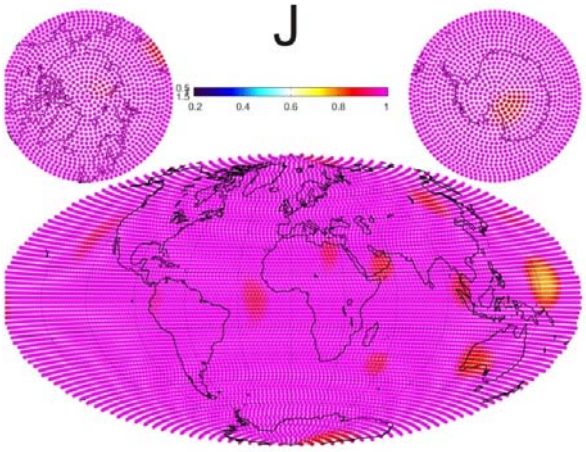


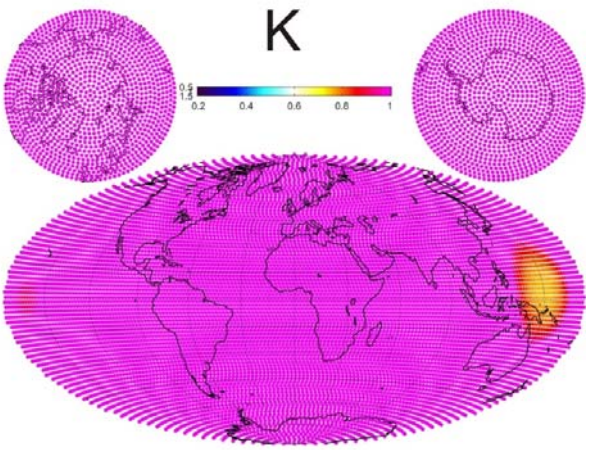
Differences in space between the median and the Huber model are in excess of 3-4 nT at the Earth's mean radius. This may be considered as negligible. However, a dipolar structure is visible.

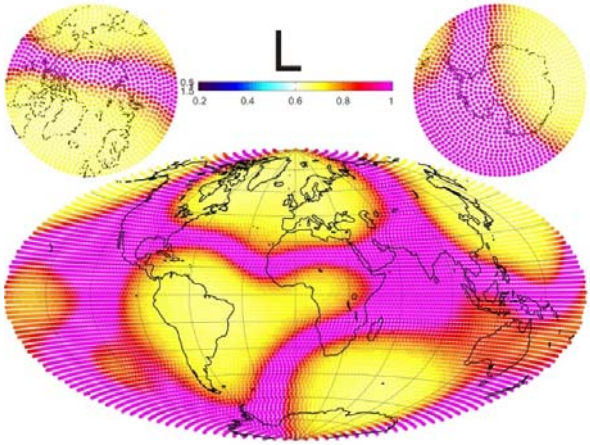
SV 2020-2025

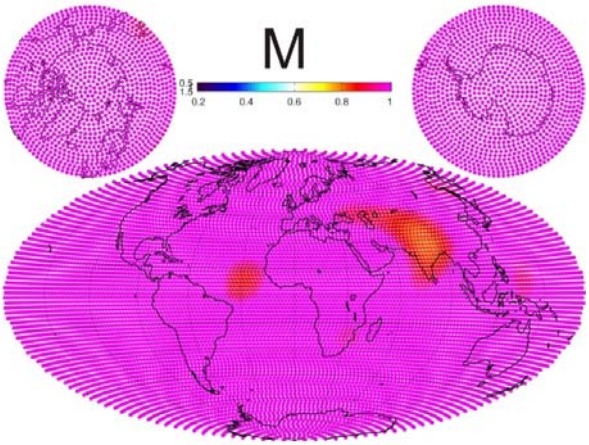
Models I, N and O show large scale differences that are downweighted by the Huber weighting scheme in space. Model H is moderatly downweighted. **The difference between the Huber weighted model and the median model in space is signficant and reaches 5nT/yr. A dipolar structure is also visible in this plot showing that the Huber and median models disagree also on the dipole coefficient.** A possible bias on the dipole coefficients could be introduced by model O when computing the median model.

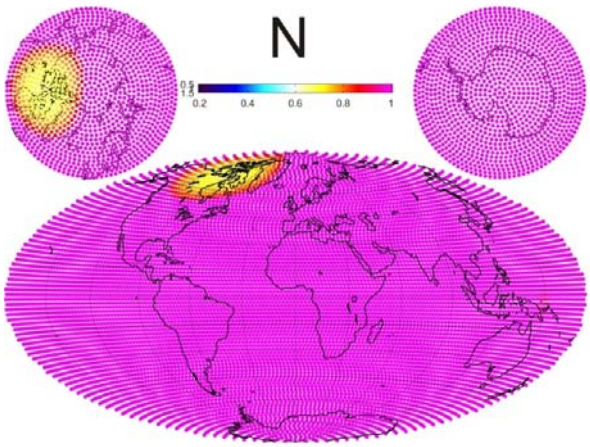

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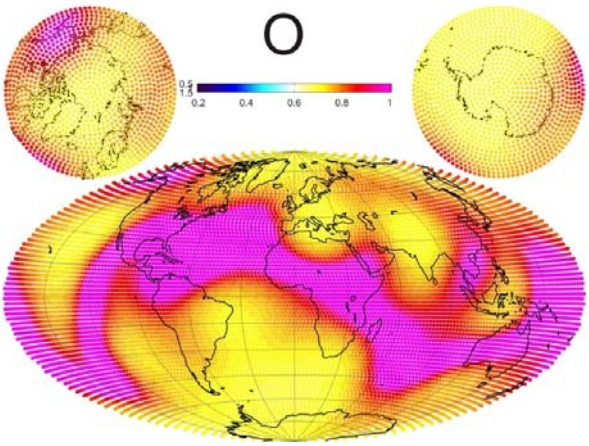

J
0.2
0.4
0.6
0.8
1


K
0.2
0.4
0.6
0.8
1


L
0.2
0.4
0.6
0.8
1


M
0.2
0.4
0.6
0.8
1


N
0.2
0.4
0.6
0.8
1


O
0.2
0.4
0.6
0.8
1

Huber weighted coefficients for SV 2020-2025

Candidate model with Huber weights in space, E. Thebault, 15-Nov-2019 13:30:49
Epoch = 2015.00

n	m	gnm	hnm
1	0	5.72	0
1	1	7.42	-25.90
2	0	-10.97	0
2	1	-6.96	-30.19
2	2	-2.14	-22.40
3	0	2.19	0
3	1	-5.87	6.01
3	2	3.06	-1.09
3	3	-12.04	0.46
4	0	-1.19	0
4	1	-1.55	-0.12
4	2	-5.93	6.48
4	3	5.20	3.55
4	4	-5.10	-4.96
5	0	-0.27	0
5	1	0.49	-0.01
5	2	-0.60	2.45
5	3	0.20	-0.55
5	4	1.29	2.98
5	5	0.85	0.28
6	0	-0.54	0
6	1	-0.34	0.04
6	2	0.41	-1.62
6	3	1.31	-1.32
6	4	-1.41	0.81
6	5	-0.02	0.02
6	6	0.91	0.97
7	0	-0.13	0
7	1	-0.15	0.61
7	2	-0.03	0.57
7	3	0.69	-0.75
7	4	0.09	-0.23
7	5	-0.54	-1.11
7	6	-0.75	0.14
7	7	0.76	0.30
8	0	-0.02	0
8	1	0.14	-0.24
8	2	-0.05	0.59
8	3	0.43	-0.17
8	4	-0.06	0.52
8	5	0.41	-0.29
8	6	0.31	-0.44
8	7	-0.08	0.49
8	8	0.39	0.03

Difference between the IGRF SV median and Huber models at the Earth's mean radius.

The difference at the Earth's mean radius is between +- 5 nT/yr in some region. This is significant. A dipole structure is also discernible.

